MTA Daily Ridership Analysis

A data analysis project that explores daily ridership and traffic across New York's transit network subways, buses, Long Island Railroad, Metro-North, Access-A-Ride, and Bridges & Tunnels since March 2020, comparing current ridership levels against pre-COVID baselines to uncover post-pandemic recovery patterns.

Tools Used

Tool	Purpose
Python	Data cleaning, preprocessing, and exploratory data analysis (EDA)
SQL	Data querying and aggregation
Power BI	Interactive dashboard and data visualization

Business Problem

Following COVID-19, the MTA — the largest transportation network in North America — experienced an unprecedented drop in ridership across all of its services. This project addresses the question: How has ridership recovered across the six MTA modes since March 2020, and which services have returned to (or remain below) pre-pandemic levels?

The goal is to give stakeholders a clear, data-driven view of post-pandemic recovery patterns across different modes of transportation to support planning and decision-making.

Data

- Source: MTA Daily Ridership and Traffic: Beginning 2020 — published by the New York Metropolitan Transportation Authority (MTA) on the NY Open Data portal.
- Coverage: Systemwide daily ridership and traffic estimates for six modes:
    - Subways (incl. Staten Island Railway)
    - Buses (NYCT & MTA Bus Company)
    - Long Island Rail Road (LIRR)
    - Metro-North Railroad (MNR)
    - Access-A-Ride Paratransit
    - Bridges & Tunnels
- Time range: From 3/1/2020 (4/1/2020 for LIRR and Metro-North) to present.
- Comparison metric: A percentage comparison against a comparable pre-pandemic date for each mode.

Methodology

Data collection (as documented by MTA):

- Subway ridership is measured from MetroCard and OMNY swipes and taps.
- Bus ridership (3/23/2020–9/30/2020) was estimated using models combining MetroCard/OMNY data and Automatic Passenger Counters; from 10/1/2020 onward it is a direct count plus cash fare estimates.
- LIRR and Metro-North ridership is modeled from monthly, weekly, ten-trip, and daily ticket sales.
- Access-A-Ride figures represent scheduled trips (a demand indicator, not completed trips).
- Bridges & Tunnels data comes from preliminary toll collection system volumes.
- Pre-pandemic baseline: each day is compared to the 2019 monthly average for the same day type (non-holiday weekday / Saturday / Sunday-holiday), per the upgraded Feb 2023 methodology.

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